

INTERNSHIP PROGRAMME

At



An Internship Report Submitted to

KERALA TECHNOLOGICAL UNIVERSITY

*In partial fulfilment of the requirements
For the award of the degree of*

BACHELOR OF TECHNOLOGY IN MECHANICAL ENGINEERING

*Submitted by
III year B tech Students of*



COLLEGE OF ENGINEERING PERUMON

*Affiliated to AICTE
Under CAPE, Established by Government of Kerala
An ISO 9001:2015 Certified Institution
Perumon P.O, Kollam 691601*

(2019-2020)

LIST OF STUDENTS

1. Abhijith S
2. Jithu Prasad
3. Ashik Nowshad
4. Praveen Prasad
5. Ravi Sankar
6. Akhil Raj
7. Rufus Rubin Oscar Fernandez
8. Rahul R
9. Vignesh M S
10. Suraj Suresh
11. Nidhin G S
12. Nandu B
13. Pranav Prakash S

TABLE OF CONTENTS

1. List of Student.....	2
2. Table of Contents.....	3
3. Acknowledgment.....	4
4. Abstract.....	5
5. Introduction.....	6
6. Objectives of the Internship.....	7
7. Scope of Internship.....	8
8. Limitations of the Internship.....	9
9. The Indian Automotive Industry.....	10
10. Company Profile.....	11
11. Quality Policy.....	12
12. Vision.....	13
13. Mission.....	13
14. Plant.....	14
15. Quality Assurance.....	16
16. Research and Development.....	17
17. Products.....	18
18. Organisational Chart.....	20
19. Organisational Hierarchy.....	21
20. Product Profile.....	22
21. Various Departments.....	23
22. The Ambitious EV Project.....	24
23. Conclusion.....	26
24. Bibliography.....	27

ACKNOWLEDGMENT

“Gratitude is not a thing of expression; it is more a matter of feeling”

We wish to express our sincere gratitude to all the staff members officials and the workers of **Kerala Automobiles Ltd (KAL)** for providing us this opportunity to do our Internship in “KERALA AUTOMOBILES LTD”.

We sincerely thank all the technicians and workers for their guidance and encouragement in carrying out this Internship. We also wish to express our gratitude to all the staffs of Kerala Automobiles Ltd who rendered their help during the period of our Internship.

We also thank the Principal of College of Engineering Perumon **Dr. Z.A. Zoya** for providing us the opportunity to embark on this Internship.

ABSTRACT

The internship enables the students to harmonize what they learnt in class with the reality in professional grounds. As a partial fulfilment for the award for a **Bachelor's Degree in Technology for Mechanical Engineering in Kerala Technical University (KTU) – College of Engineering Perumon (CEP)**, it is very fundamental for any student in his/her learning period to undertake practical training. We took our internship programme at **KERALA AUTOMOBILES LTD (KAL), Aralumood P.O Neyyattinkara Thiruvananthapuram**. Our Internship started on 24th July 2019 and ended on 30th July 2019. We were eager to know how this Automobile Company fulfils its obligations and how it operates and, we took advantage of the internship and found different aspects and stages of manufacturing of an Auto rickshaw and the manufacturing techniques of different parts for the **ISRO** and **VSSC**.

INTRODUCTION

The main aim of this Internship is familiarization and in depth knowledge of the functioning of various departments. For this purpose an Internship has been carried out in **KERALA AUTOMOBILES LTD.** The KAL has a decent position in the auto industry which manufactures three wheelers with high power, fuel efficiency, and strong chassis, metal body, and high load carrying capacity, with trouble free operation. Also the latest addition the e-Auto by KAL is also a sign of a new wave development in Automotive Trend which the Kerala Automobiles Ltd is following. Even though the company is on loss now according to prediction it will be having great success in the coming years because of the e-Auto.

OBJECTIVES OF THE INTERNSHIP

- Getting Industrial exposure.
- To see and understand the actual working condition of an Automobile Company.
- To understand the functions of each and every department.
- To look into the working conditions of the employees in a Company.
- To make the evaluation about the performance of the Company.
- To know about the Company objectives and scope.
- To know the extent to which management theories matches with the practice.
- To suggest recommendations for improving the overall performance of the Company.
- To get clear view of the day to day functioning of the industrial establishment.

SCOPE OF THE INTERNSHIP

- To understand the working environment.
- To understand the employee-worker relationship
- To understand the profit making of the Company.
- How man, material, money influences on the achievement of company.
- To learn the HR functions of the Company.
- To learn the Industrial Relations functions in the company.

LIMITATIONS OF THE INTERNSHIP

- Time was limited for a deep analysis of the Company.
- Some company documents are confidential in nature. So, it was not possible to include all those items.
- Industrial environment was very busy, which limits the data collection, learning. All the information couldn't be included because of this factor.

THE INDAIN AUTOMOTIVE INDUSTRY

The Indian Automobile industry is the fourth largest in the world with an annual production of over 4.02 million units in 2017 (excluding two wheelers). In 2018 India emerged as the seventh largest exporter of automobiles, behind Taiwan, Singapore, Hong Kong, South Korea, Japan and China. Following economic liberalization in 1991, the Indian Automotive industry has demonstrated sustained growth as a result of increased competitiveness and relaxed restrictions. Several Indian automobile manufacturers such as TATA Motors, Maruthi Suzuki, Mahindra & Mahindra, expanded their domestic and international operations. India's robust economic growth led to the further expansion of its domestic automobile market which attracted significant India-specific investment by multinational automobile manufacturers. In 2017, the motor vehicle sales were 4 Million units.

Embryonic automotive industry emerged in India in the 1940's. Following the independence, in 1947, the government of India and the private sector launched efforts to create an automotive manufacturing industry to supply to the automotive industry. However, the growth was relatively slow in the 1950's and 1960's due to nationalization and the licence raj which hampered the Indian Private Sector. After 1970's, the automotive industry started to grow, but the growth was mainly driven by tractors, commercial vehicles and scooters. Cars were still a major luxury. Japanese Manufacturers entered the Indian Market ultimately leading to the establishment of Maruthi Udyog. A number of foreign firms initiated joint ventures with Indian Companies.

In the 1980's a number of Japanese Manufacturers launched joint-ventures for building a light commercial vehicle. It was this time that Indian government chose Suzuki for this joint-venture to manufacture small cars. Following the economic liberalisation in 1991 and the gradual weakening of the licence raj, a number of Indian and multinational car companies launched operations. Since then, automotive component and automobile manufacturing growth has accelerated to meet domestic and export demands.

COMPANY PROFILE

KERALA AUTOMOBILES LIMITED (KAL) incorporated in 1978 as a Government of Kerala undertaking is set up in the picturesque back drop, 16km south of Thiruvananthapuram in a tiny village called *Aralummood*, in Neyyattinkara Taluk. The Company manufactures **Three Wheelers** (*Diesel, Petrol, CNG and Electric*) suitable for passengers and goods traffic in the brand name of **KERALA** and are considered as most Eco-friendly vehicles.

KAL started production with 164 employees where capacity of production was 52 vehicles per year. Initial technology of production was in collaboration with API Mumbai and from 1987 it has developed its own technology of production.

KAL started with a production of petrol engine three wheelers from 1988 it began to produce 'diesel engine' three wheelers. KAL is the first company to introduce diesel engine three wheelers in the country. At present the company is concentrating in the diesel versions and Electric versions only.

The strong, highly motivated work force in the Technical and Management cadres are working hand in hand for the progress of the Company. The ancillary units developed by the Company around the factory complex provide livelihood for hundreds of families.

Since commencement of production in 1984, the Company has manufactured and marketed more than 1,00,000 Three Wheelers. KAL has distinguished itself as a force worth reckoning in the Automobile industry in India. The Company has also exported a number of Three Wheelers to Bangladesh, Sri Lanka, Nepal, Sudan, Botswana, Nigeria, South Africa, Madagascar; and Guatemala (Central America). the Company could turn the corner for the first time in 1993-94 and continued to keep the same trend at a higher level. The acceptability of KAL Three Wheelers in the Third World countries speaks for its utility and quality. KAL is also manufacturing sophisticated components to be used in various space programs of ISRO (VSSC, LPSC, IISU)

The Company is getting all kind of support from the Government of Kerala, without which it would not have been possible to achieve the present level of promising state of affairs. KAL was awarded with ISO Certification in 1998 itself and keep hold of the same ever since by updating the quality management system appropriately to conform to the amendments in the ISO Standards. Currently KAL is certified as ISO 9001:2008 compliant in recognition to its strong customer focus, the motivation and implication of top management, the process approach and continual improvement. It is the unflinching faith of our customers, which has powered KAL into the path of progress and looking forward for continued patronage at a higher rank.

QUALITY POLICY

“The Management and Employees of KAL are committed to develop, manufacture, market and service three wheelers, enhancing customer satisfaction through a process of continual improvement of its Quality Management System”



VISION

"To become top three wheeler manufacturer in 10 years"

MISSION

"To provide value added service and product to the customers and to earn reasonable profit"

PLANT

Kerala Automobiles Limited (KAL)'s state-of-the-art factory is equipped with the most modern CNC machines to ensure dimensional accuracy of each component with reference to the specification requirements. The Factory has highly sophisticated testing facilities to maintain stringent quality control at every stage of production. The proof of the high quality standards maintained can be apprehended from the fact that KAL has the distinctive privilege of supplying critical components to the Indian Space Research Organisation (ISRO), for India's sterling space programs.





QUALITY ASSUARANCE

At every stage of manufacturing, excellent quality assurance is **KAL**'s prime objective. This has been made possible using equipments such as Universal Testing Machines, Dynamometers, Exhaust Analyser, Digital Height Master, Ultrasonic Crack Detector, Eddy Current Tester etc.

Quality assurance achieved through Total Quality Management and up to date Quality management system has helped **KAL** to get **ISO 9001:2008 Certification**.



RESEARCH AND DEVELOPMENT

KAL's Product Engineering Department has computerised design systems and testing equipment to ensure international standards. Experienced engineers have transformed the basic Italian reference to the latest evolution of the design leading to fuel efficient Eco-friendly **KAL** Three Wheelers ensuring stimulating quality system.

The works of various new models are in progress.

PRODUCTS

KAL product range comprises of **Three Wheelers** (Diesel, Petrol & CNG) in the brand name of **KERALA**, suitable for passengers and goods traffic

KERALA RE V.go DIESEL



KAL RE V.go Diesel Auto rickshaw(Fully built)

Vehicle Specification	VEHICLE DESCRIPTION	
	Diesel Model	CNG Model
Overall Length (mm)	2920	NA
Overall Width (mm)	1350	NA
Wheel Base (mm)	1890	NA
G V W (Kg)	850	NA
Tyre	4.5" X 10" (8Ply)	NA
Fuel Tank Capacity (Ltr)	10.5	NA
Fuel consumption (Kmpl)	37	NA



Long Chassis (MX-400)

KAL Chassis available for Passenger and Goods carrier in Diesel model.

VEHICLE DESCRIPTION		
Vehicle Specification	Diesel Model	CNG Model
Overall Length	3480 mm	
Overall Width	1310 mm	
Wheel Base	2140 mm	
Ground Clearance	120 mm	
GVW	930 Kg	

Tipper



KAL Tipper available in *Diesel* model.

VEHICLE DESCRIPTION

Vehicle Specification	Diesel Model	CNG Model
Overall Length	2515 mm	
Overall Width	1375 mm	
Wheel Base	1870 mm	
Ground Clearance	120 mm	
GVW	750 Kg	

ORGANISATIONAL CHART

The Board of Directors manages the affairs of the company. The Government of Kerala determines the members of the board. The members are determined by the Government in a conservative way. The number shall not be less than two and not more than nine. The Government appoints the director including the full time directors. The directors need not hold shares. They frame policies requires for the company to achieve its goals. The managing director will execute the affairs of the company through the managers in different departments. It consists of

- I. Chairman
- II. Managing Directors
- III. A Nominee from Finance Department
- IV. A Nominee from Industries Department
- V. A Nominee from the Government

The MD is always responsible for the conduct of managers of the company subject to overall and supervision by board of directors. The general shall be in charge of the company where MD is not present. In such cases, he or she shall take decision on financial matters in conclusion with financial controllers.

ORGANISATIONAL HIERARCHY



PRODUCT PROFILE

Petrol Models

- P1.** Petrol Auto Rickshaw, Regular three seater (1984)
- P2.** New look Auto Rickshaw (1988)
- P3.** Petrol Delivery Van (1984)
- P4.** Petrol Pickup Van (1984)
- P5.** Petrol three Wheeler Chassis (1985)

Diesel Models

- D1.** Diesel Pickup Van (1984)
- D2.** Diesel Auto Trailers (1988)
- D3.** Diesel Chassis come under three models
 - 1. 4360 Chassis- Self Start
 - 2. 4325 Shorter rope Start
- D4.** Kerala MX 4000
- D5.** LGA 340 (ELECTRIC START)

Current Moving Products

Kerala MX 4000

Kerala V2 Diesel Chassis (Self & Electric)

VARIOUS DEPARTMENTS

There are approximately 20 Departments in KAL. The size of the departments depends on the importance and complexity of the function of the department.

- 1) Human Resource Department
- 2) Administrative Department
- 3) Sub-Contract Department
- 4) Research and Development Department
- 5) Purchase Department
- 6) Finance Department
- 7) Marketing Department
- 8) Production Department
- 9) Planning Department
- 10) Machine Shop Department
- 11) Quality Assurance Department
- 12) Stores Department
- 13) Maintenance Department
- 14) Service and Spare Parts

THE AMBITIOUS EV PROJECT



Kerala Automobiles Limited (KAL) will begin the production of electric auto rickshaws from July 10th 2019 with an aim to launch 8000 vehicles in the first phase.

KAL will also enter into the production of electric buses within next nine months in collaborations with Swiss electric bus manufacturer **HESS AG**

Automotive Research Association of India has already given certification to KAL for manufacturing electric auto rickshaws. KAL will be soon accepting orders from the market for auto rickshaw.

The e-Autos will be capable of travelling 100km after full recharge. The mileage can be increased if the vehicle is charged when it is at rest.

The Lithium-based batteries can be charged using ordinary plug sockets at home. They will also launch electric charging stations across the state with help from **Kerala Electricity Board (KAL)**. e-Autos, which will cost around Rs. 2.5 Lakh per unit, can be operated at a cost of 50 paise per km. Thus it will fetch better profits for the auto rickshaw drivers. Setting up of charging stations at auto stands is also on cards. Maintenance cost will also be lower than petrol/diesel auto rickshaws.

Swiss company Hess AG had evinced interest in setting up an electric bus manufacturing unit in the state after learning about the state government's electric vehicle policy and the plans to switch over to electric vehicles soon.

The state government had last year framed and announced an electric vehicle policy that targets one million electric vehicles on the roads in the state by 2022. This also includes 3000 electric buses. In

the electric vehicle policy, buses will be given more preference, due to its potential to reduce pollution and promote shared mobility.

E-buses for public transport are envisaged with an average driving range of 50km to 100km in one recharge.

The policy also envisages that a part of KSRTC's fleet of 6000 buses will be converted to electric buses by 2025 with financial aid from centre. The KSRTC, on an average, procures around 1000 buses annually. Switching over to electric buses also reduces maintenance and fuel costs.

Replacing diesel-run buses with e-buses which cause neither noise nor air pollution may change the face of public transport in Kerala. They have plenty of other advantages too, like rock-bottom operational expenses due to low cost of energy (electricity).

Wear and tear too is low since e-buses have only around 20% of moving parts as compared to their conventional counterparts. However, the biggest concern is that they are priced much higher. An e-bus costs Rs 1.50 Crore unlike conventional buses that cost between Rs 35 lakh to RS 40 lakh.

KAL's initiative is part of the target of putting one million EV's on the road by 2022, with a pilot fleet of 200000 two-wheelers, 50000 three-wheelers, 1000 goods carriers, 3000 buses and 100 ferry boats by 2020.



CONCLUSION

KAL is one of the first three wheelers manufacturers of India. The products of KAL are sold under the brand name "KAL". KAL was established in 15th March 1978 and was officially inaugurated in 1984. KAL is a public sector company. As that of any public sector enterprises KAL's gestation period was long with many ups and down. But a committed skilful work force and efficient teams of management has helped KAL to overcome all the obstacles. Company has already entered into the international market and it has strong hold in markets in North Indian states with a slight improvement in technological aspects and challenge to its competitors like Bajaj Auto Ltd., Piaggio, and TVS etc. The introduction of e-Auto is a change in the trend of Automotive Industry which is a sign of progress for the future of KAL. The only drawback is that presently the company is not able to meet its demanded production due to inadequate resources

BIBLIOGRAPHY

- ❖ Company Journals
- ❖ Department Files
- ❖ Direct Interview
- ❖ www.threewheelersindia.com
- ❖ www.keralaautomobiles.com